

Untitled

2026-10-01

R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

```
library(janitor)
```

```
##
## Attaching package: 'janitor'

## The following objects are masked from 'package:stats':
##
##   chisq.test, fisher.test
```

```
library(tidyverse)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.2
```

```
## Warning: package 'tibble' was built under R version 4.5.2
```

```
## Warning: package 'tidyr' was built under R version 4.5.2
```

```
## Warning: package 'readr' was built under R version 4.5.2
```

```
## Warning: package 'purrr' was built under R version 4.5.2
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.6
## v forcats    1.0.1      v stringr    1.6.0
## v ggplot2    4.0.1      v tibble     3.3.1
## v lubridate  1.9.4      v tidyr      1.3.2
## v purrr      1.2.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag() masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)
```

```
rats <- read.csv("Rat_sightings.csv")
head(rats)
```

```
## Unique.Key Created.Date Closed.Date
## 1 43244587 07/10/2019 11:34:15 AM 11/25/2019 02:24:48 PM
## 2 39866973 07/29/2018 12:40:10 PM 08/02/2018 04:29:49 PM
## 3 39866175 07/29/2018 05:01:59 AM 08/02/2018 12:00:00 AM
## 4 39866174 07/29/2018 02:05:47 PM 08/02/2018 03:21:31 PM
## 5 39865617 07/29/2018 01:23:25 PM 08/01/2018 01:28:30 PM
## 6 39863596 07/29/2018 07:43:04 PM 07/29/2018 07:43:04 PM
## Location.Type Incident.Zip Incident.Address
## 1 3+ Family Apt. Building 11356 114-35 TAIPEI COURT
## 2 1-2 Family Dwelling 11368 102-17 CORONA AVENUE
## 3 3+ Family Apt. Building 10024 315 WEST 88 STREET
## 4 Other (Explain Below) 11236 1028 EAST 81 STREET
## 5 3+ Family Mixed Use Building 10026 130 WEST 113 STREET
## 6 1-2 Family Dwelling 11236 629 EAST 89 STREET
## Street.Name City Status Community.Board Borough Latitude
## 1 TAIPEI COURT College Point Closed 07 QUEENS QUEENS 40.78873
## 2 CORONA AVENUE Corona Closed 04 QUEENS QUEENS 40.74425
## 3 WEST 88 STREET NEW YORK Closed 07 MANHATTAN MANHATTAN 40.79043
## 4 EAST 81 STREET BROOKLYN Closed 18 BROOKLYN BROOKLYN 40.63378
## 5 WEST 113 STREET NEW YORK Closed 10 MANHATTAN MANHATTAN 40.80080
## 6 EAST 89 STREET BROOKLYN Closed 18 BROOKLYN BROOKLYN 40.64207
## Longitude Descriptor Complaint.Type
## 1 -73.85363 Rat Sighting Rodent
## 2 -73.86127 Rat Sighting Rodent
## 3 -73.97746 Rat Sighting Rodent
## 4 -73.91004 Rat Sighting Rodent
## 5 -73.95281 Rat Sighting Rodent
## 6 -73.90932 Rat Sighting Rodent
```

```
glimpse(rats)
```

```
## Rows: 290,641
## Columns: 15
## $ Unique.Key <int> 43244587, 39866973, 39866175, 39866174, 39865617, 398~
## $ Created.Date <chr> "07/10/2019 11:34:15 AM", "07/29/2018 12:40:10 PM", "~
## $ Closed.Date <chr> "11/25/2019 02:24:48 PM", "08/02/2018 04:29:49 PM", "~
## $ Location.Type <chr> "3+ Family Apt. Building", "1-2 Family Dwelling", "3+~
## $ Incident.Zip <chr> "11356", "11368", "10024", "11236", "10026", "11236",~
## $ Incident.Address <chr> "114-35 TAIPEI COURT", "102-17 CORONA AVENUE", "315 W~
## $ Street.Name <chr> "TAIPEI COURT", "CORONA AVENUE", "WEST 88 STREET", ~
## $ City <chr> "College Point", "Corona", "NEW YORK", "BROOKLYN", "N~
## $ Status <chr> "Closed", "Closed", "Closed", "Closed", "Closed", "Cl~
## $ Community.Board <chr> "07 QUEENS", "04 QUEENS", "07 MANHATTAN", "18 BROOKLY~
## $ Borough <chr> "QUEENS", "QUEENS", "MANHATTAN", "BROOKLYN", "MANHATT~
## $ Latitude <dbl> 40.78873, 40.74425, 40.79043, 40.63378, 40.80080, 40.~
## $ Longitude <dbl> -73.85363, -73.86127, -73.97746, -73.91004, -73.95281~
## $ Descriptor <chr> "Rat Sighting", "Rat Sighting", "Rat Sighting", "Rat ~
## $ Complaint.Type <chr> "Rodent", "Rodent", "Rodent", "Rodent", "Rodent", "Ro~
```

```
names(rats)
```

```
## [1] "Unique.Key"      "Created.Date"    "Closed.Date"     "Location.Type"
## [5] "Incident.Zip"    "Incident.Address" "Street.Name"     "City"
## [9] "Status"          "Community.Board" "Borough"         "Latitude"
## [13] "Longitude"       "Descriptor"      "Complaint.Type"
```

```
dim(rats)
```

```
## [1] 290641    15
```

```
colSums(is.na(rats))
```

```
##      Unique.Key      Created.Date      Closed.Date      Location.Type
##           0           0           0           0
## Incident.Zip Incident.Address      Street.Name           City
##           0           0           0           0
##      Status Community.Board      Borough      Latitude
##           0           0           0      2272
##      Longitude      Descriptor      Complaint.Type
##      2272           0           0
```

```
# Load and clean
rats <- read_csv(
  "Rat_sightings.csv",
  na = c("", "NA"),
  col_types = cols(
    `Incident Zip` = col_character(),
    .default = col_guess()
  )
) %>%
  clean_names() %>%
  distinct(unique_key, .keep_all = TRUE) %>%
  mutate(
    # Convert dates
    created_date = mdy_hms(created_date),
    closed_date = mdy_hms(closed_date),

    # Standardize text
    borough = str_to_title(str_trim(borough)),
    location_type = str_squish(location_type),
    descriptor = str_squish(descriptor),

    # Extract useful time variables
    year = year(created_date),
    month = month(created_date, label = TRUE, abbr = TRUE),
    month_num = month(created_date)
  ) %>%
  filter(!is.na(created_date))

# Check cleaned data
glimpse(rats)
```

```
## Rows: 290,641
## Columns: 18
## $ unique_key      <dbl> 43244587, 39866973, 39866175, 39866174, 39865617, 398~
## $ created_date    <dtm> 2019-07-10 11:34:15, 2018-07-29 12:40:10, 2018-07-29~
## $ closed_date     <dtm> 2019-11-25 14:24:48, 2018-08-02 16:29:49, 2018-08-02~
## $ location_type   <chr> "3+ Family Apt. Building", "1-2 Family Dwelling", "3+~
## $ incident_zip    <chr> "11356", "11368", "10024", "11236", "10026", "11236",~
## $ incident_address <chr> "114-35 TAIPEI COURT", "102-17 CORONA AVENUE", "315 W~
## $ street_name     <chr> "TAIPEI COURT", "CORONA AVENUE", "WEST 88 STREET", ~
## $ city            <chr> "College Point", "Corona", "NEW YORK", "BROOKLYN", "N~
## $ status          <chr> "Closed", "Closed", "Closed", "Closed", "Closed", "Cl~
## $ community_board <chr> "07 QUEENS", "04 QUEENS", "07 MANHATTAN", "18 BROOKLY~
## $ borough         <chr> "Queens", "Queens", "Manhattan", "Brooklyn", "Manhatt~
## $ latitude        <dbl> 40.78873, 40.74425, 40.79043, 40.63378, 40.80080, 40.~
## $ longitude       <dbl> -73.85363, -73.86127, -73.97746, -73.91004, -73.95281~
## $ descriptor      <chr> "Rat Sighting", "Rat Sighting", "Rat Sighting", "Rat ~
## $ complaint_type  <chr> "Rodent", "Rodent", "Rodent", "Rodent", "Rodent", "Ro~
## $ year            <dbl> 2019, 2018, 2018, 2018, 2018, 2018, 2018, 2018, 2018, 2018,~
## $ month           <ord> Jul, Jul, Jul, Jul, Jul, Jul, Jul, Jul, Jul, Jul, Jul, Jul,~
## $ month_num       <dbl> 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7,~
```

```
summary(rats)
```

```
##      unique_key      created_date      closed_date
## Min.   :11464394   Min.   :2010-01-01 08:29:58   Min.   :1899-12-31 19:00:00
## 1st Qu.:32884157   1st Qu.:2016-03-11 00:00:00   1st Qu.:2016-05-27 00:00:00
## Median :46184201   Median :2020-05-12 13:38:39   Median :2020-06-19 08:49:45
## Mean   :45001590   Mean   :2019-08-02 03:57:51   Mean   :2019-10-02 14:31:19
## 3rd Qu.:57669622   3rd Qu.:2023-05-21 16:56:15   3rd Qu.:2023-06-23 14:21:45
## Max.   :70160809   Max.   :2026-08-23 01:13:58   Max.   :2026-08-21 20:00:25
##                                     NA's   :17776
## location_type      incident_zip      incident_address      street_name
## Length:290641      Length:290641      Length:290641      Length:290641
## Class :character    Class :character    Class :character    Class :character
## Mode  :character    Mode  :character    Mode  :character    Mode  :character
##
##
##
##      city      status      community_board      borough
## Length:290641   Length:290641   Length:290641   Length:290641
## Class :character Class :character Class :character Class :character
## Mode  :character Mode  :character Mode  :character Mode  :character
##
##
##
##      latitude      longitude      descriptor      complaint_type
## Min.   :40.11      Min.   : -77.52      Length:290641      Length:290641
## 1st Qu.:40.68      1st Qu.: -73.97      Class :character    Class :character
## Median :40.72      Median : -73.94      Mode  :character    Mode  :character
## Mean   :40.74      Mean   : -73.94
## 3rd Qu.:40.81      3rd Qu.: -73.90
## Max.   :40.91      Max.   : -73.70
```

```
## NA's :2272 NA's :2272
## year month month_num
## Min. :2010 Jul : 32610 Min. : 1.000
## 1st Qu.:2016 Aug : 31881 1st Qu.: 4.000
## Median :2020 Jun : 31189 Median : 7.000
## Mean :2019 May : 30385 Mean : 6.518
## 3rd Qu.:2023 Sep : 26962 3rd Qu.: 9.000
## Max. :2026 Apr : 24623 Max. :12.000
## (Other):112991
```

```
# Missing values
colSums(is.na(rats))
```

```
## unique_key created_date closed_date location_type
## 0 0 17776 2
## incident_zip incident_address street_name city
## 427 10933 10934 4036
## status community_board borough latitude
## 0 2 2 2272
## longitude descriptor complaint_type year
## 2272 0 0 0
## month month_num
## 0 0
```

```
# Borough counts
rats %>%
  count(borough, sort = TRUE)
```

```
## # A tibble: 7 x 2
## borough n
## <chr> <int>
## 1 Brooklyn 107028
## 2 Manhattan 75729
## 3 Bronx 51559
## 4 Queens 45333
## 5 Staten Island 10972
## 6 Unspecified 18
## 7 <NA> 2
```

```
# Years represented
rats %>%
  count(year, sort = TRUE)
```

```
## # A tibble: 17 x 2
## year n
## <dbl> <int>
## 1 2022 26500
## 2 2023 25452
## 3 2024 25191
## 4 2021 25012
## 5 2025 19896
## 6 2017 19152
```

```
## 7 2018 17353
## 8 2016 17230
## 9 2019 16808
## 10 2020 16262
## 11 2015 15286
## 12 2014 12617
## 13 2026 11512
## 14 2013 10739
## 15 2012 10643
## 16 2010 10534
## 17 2011 10454
```

```
# Location types
rats %>%
  count(location_type, sort = TRUE)
```

```
## # A tibble: 56 x 2
##   location_type      n
##   <chr>          <int>
## 1 3+ Family Apt. Building 125069
## 2 1-2 Family Dwelling   53577
## 3 Other (Explain Below) 42627
## 4 3+ Family Mixed Use Building 19133
## 5 Commercial Building  15030
## 6 Vacant Lot           7183
## 7 Construction Site    5874
## 8 1-2 Family Mixed Use Building 4723
## 9 Vacant Building      4284
## 10 Parking Lot/Garage   2966
## # i 46 more rows
```

```
# Date range
range(rats$created_date, na.rm = TRUE)
```

```
## [1] "2010-01-01 08:29:58 UTC" "2026-08-23 01:13:58 UTC"
```

```
monthly_rats <- rats %>%
  filter(
    borough %in% c(
      "Manhattan", "Brooklyn", "Queens",
      "Bronx", "Staten Island"
    )
  ) %>%
  group_by(borough, year, month_num) %>%
  summarize(
    sightings = n(),
    .groups = "drop"
  ) %>%
  mutate(
    date = as.Date(
      paste(year, month_num, "01", sep = "-")
    )
  )
```

```

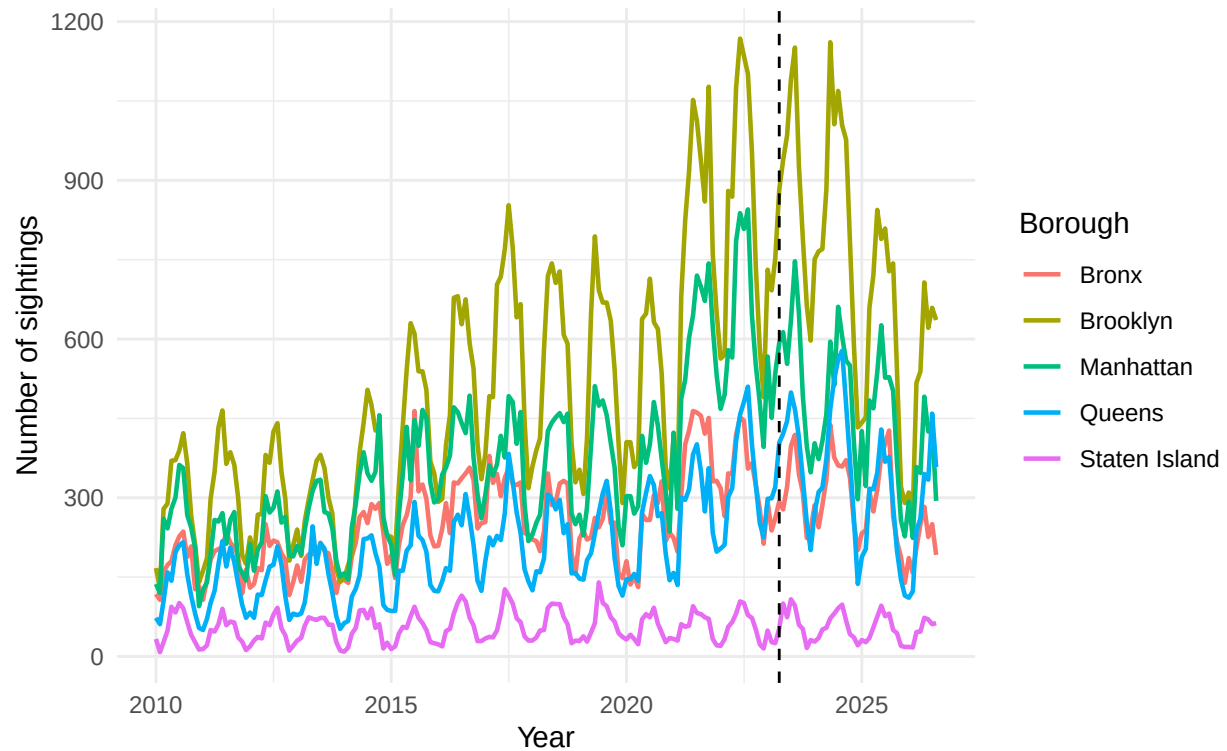
)

# View the summary
view(monthly_rats)

ggplot(
  monthly_rats,
  aes(
    x = date,
    y = sightings,
    color = borough
  )
) +
  geom_line(linewidth = 0.8) +
  geom_vline(
    xintercept = as.Date("2023-04-01"),
    linetype = "dashed",
    color = "black"
  ) +
  labs(
    title = "Monthly Rat Sightings by Borough",
    subtitle = "NYC 311 complaints, with April 2023 marked",
    x = "Year",
    y = "Number of sightings",
    color = "Borough"
  ) +
  theme_minimal()

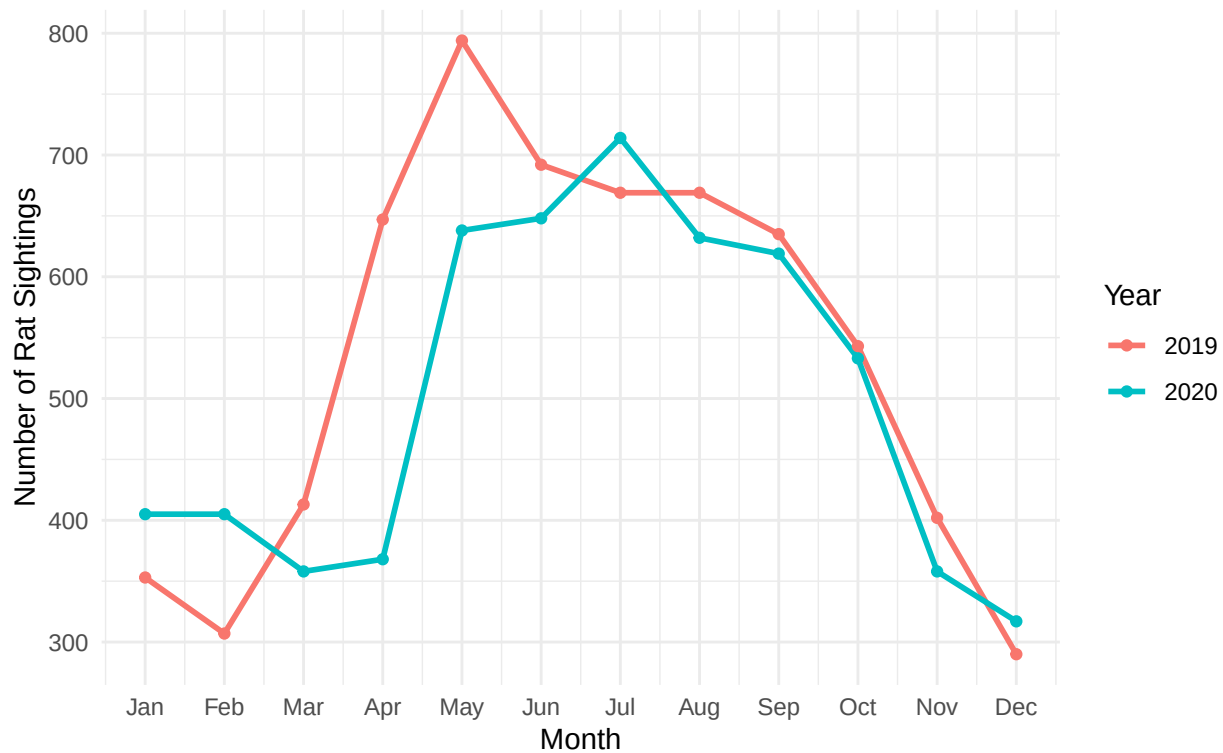
```

Monthly Rat Sightings by Borough
 NYC 311 complaints, with April 2023 marked



```
rats %>%
  filter(
    borough == "Brooklyn",
    year %in% c(2019, 2020)
  ) %>%
  count(year, month_num, month) %>%
  ggplot(aes(x = month_num, y = n, color = factor(year), group = year)) +
  geom_line(linewidth = 1) +
  geom_point() +
  scale_x_continuous(
    breaks = 1:12,
    labels = month.abb
  ) +
  labs(
    title = "Brooklyn Rat Sightings by Month",
    subtitle = "2019 vs. 2020",
    x = "Month",
    y = "Number of Rat Sightings",
    color = "Year"
  ) +
  theme_minimal()
```

Brooklyn Rat Sightings by Month
2019 vs. 2020



The graph above compares the two years with the most similar patterns of rat sightings in Brooklyn, the borough with the highest number of reported sightings. A clear seasonal pattern appears, with sightings increasing during the spring and summer before declining in the fall and winter. This pattern is consistent with research on New York City rat reports, which has found a significant relationship between warmer temperatures and increased reported rat sightings. Warmer weather may increase rat activity and visibility while also increasing the amount of time people spend outdoors, creating more opportunities for rats to be seen and reported.

However, these results should be interpreted as changes in reported sightings, not necessarily changes in the rat population. Human activity, garbage, and other features of the urban environment can also influence where and how often rats are reported. This distinction is important because a decline in 311 reports during colder months does not necessarily mean that the rat population has declined by the same amount.

The seasonal pattern has a potential policy implication for the city. Rather than waiting for complaints to increase, sanitation and rat-control efforts could be increased before the spring and summer months, when reported sightings begin to rise. Using seasonal 311 trends to anticipate periods of increased activity could help the city target resources more efficiently.